

# NetraJyoti AI

A Bengali-first, safety-guidance and eye-care navigation MVP for rural West Bengal.

| Submission information       | Value                                                             |
|------------------------------|-------------------------------------------------------------------|
| Team name                    | NETRAJYOTI                                                        |
| Team members and student IDs | Rahul Kar (bitsom_pm_2601248)<br>Subhra Kundu (bitsom_pm_2601245) |
| Course                       | Product Management with Generative and Agentic AI                 |
| Phase                        | Phase 2                                                           |
| Product                      | NetraJyoti AI MVP                                                 |

## 1. Product strategy and solution design

### 1.1 Problem statement

In rural West Bengal, Bengali-speaking users may not know whether an eye problem is serious, where to seek help, or which advice to trust. Informal advice, forwarded messages, cost, travel, waiting time, low digital confidence, and communication barriers can delay access to appropriate eye care. NetraJyoti provides Bengali-first, non-diagnostic safe next-step guidance and care navigation.

### 1.2 Design-thinking process

| Stage     | Application to NetraJyoti                                                                                                                  | Output                                    |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| Empathise | Interview rural users, caregivers, ASHA workers, and outreach coordinators about eye-care decisions, language needs, and service barriers. | Evidence-backed needs and quotes.         |
| Define    | Frame the problem as delay, confusion, and lack of trusted Bengali guidance before care.                                                   | Problem statement and jobs-to-be-done.    |
| Ideate    | Explore voice, text, human-support, service-navigation, family-sharing, and safety-routing concepts.                                       | Prioritised solution ideas.               |
| Prototype | Create Bengali-first wireframes with urgent, routine, and human-support paths.                                                             | Interactive wireframe and user flow.      |
| Test      | Run task-based usability and content-comprehension sessions; test safety rules and fallbacks.                                              | Insights, refinements, and release gates. |

### 1.3 Personas and jobs-to-be-done

| Persona                            | Need / job                                                                     | Design implication                                                |
|------------------------------------|--------------------------------------------------------------------------------|-------------------------------------------------------------------|
| Maya - rural Bengali-speaking user | Understand what to do next when an eye concern creates fear or uncertainty.    | Bengali-first, simple choices, large actions, no diagnosis claim. |
| Rahim - caregiver                  | Help a family member reach the right eye-care service and communicate urgency. | Selected symptoms, family message, share/copy, service direction. |
| Soma -                             | Support a resident who needs trusted local                                     | Human-support route, PHC/ASHA options, clear                      |

| Persona                      | Need / job                                                         | Design implication                                                                                                     |
|------------------------------|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| ASHA/community-health worker | guidance.                                                          | approved language.                                                                                                     |
| Outreach coordinator         | Keep camps, vision centres, clinics, and referral details current. | Future pilot: authenticated partner service-directory update workflow, named verification ownership, and review dates. |

## 1.4 User stories, MVP, and prioritization

| User story / safety requirement                                                                                                               | Related delivery scope                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| As a Bengali-speaking user, I want to choose my eye symptoms in Bengali so that I can understand a safe next step.                            | Consent, fixed symptom choices, urgent warning-sign check, deterministic routing.                                                                                      |
| As a caregiver, I want to share selected symptoms and approved guidance so that my family can act together.                                   | Caregiver summary, share/copy fallback, service-direction support.                                                                                                     |
| As an ASHA or community-health worker, I want a reviewed human-support route so that I can assist a resident safely.                          | Human-support path, PHC/ASHA guidance, clear escalation actions.                                                                                                       |
| As an outreach coordinator, I want to update verified service details so that users receive current eye-care information.                     | Future pilot capability: authenticated service-directory update workflow, verification ownership, and data freshness controls. The current MVP uses static demo cards. |
| Safety requirement: Deterministic reviewed rules must set the route before any AI summary is generated, so AI cannot change safety decisions. | Rules-first routing, controlled-AI feature flag, validation, approved fallback.                                                                                        |

| Priority | MVP capability                                                                                                                                              | Why                                                    |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| Must     | Bengali consent, fixed symptom selection, urgent warning check, deterministic reviewed route.                                                               | Core safety and comprehension.                         |
| Must     | Urgent care advice, routine guidance, human-support path, return/restart actions.                                                                           | No-dead-end safe navigation.                           |
| Must     | District/town or block service direction, PHC/ASHA options, family sharing.                                                                                 | Transforms guidance into practical action.             |
| Should   | Optional browser speech input with visible text fallback; patient-detail consent for share messages. Result-page audio playback is not in the current MVP.  | Improves accessibility without blocking the flow.      |
| Should   | Optional local-Ollama Bengali route explanation after Java fixes the route, with validation and fixed fallback. AI caregiver summarisation is future scope. | Adds clarity only after deterministic route selection. |
| Could    | Live service availability, richer partner integrations, appointment booking.                                                                                | Post-MVP; needs operational data governance.           |

## 1.5 Build approach, testing, and AI

- Design and core development are handled internally to retain product knowledge, Bengali UX control, safety-rule ownership, and fast iteration. Specialist clinical/content review and local service-data verification are partner-supported.
- Technology capability: React + TypeScript + Vite frontend/PWA; Java 21 + Spring Boot REST API; local PostgreSQL; Spring Security configuration; JUnit; and available Vitest/Playwright tooling. The routing and AI demo endpoints are publicly permitted in the current MVP.
- Current automated evidence is limited to Java unit tests for safety routing and output-policy validation. Before any pilot, add and record API integration, frontend, end-to-end, accessibility, usability, content/clinical-review, privacy, and failure-mode test evidence.

- AI/GenAI is optional and backend-only. Java retrieves matching PostgreSQL criteria and fixes URGENT, HUMAN SUPPORT, or ROUTINE before PostgreSQL retrieves route guidance. Local Ollama may then create a Bengali route explanation; Java validates it and returns fixed Bengali fallback copy if retrieval, generation, or validation fails. Ollama cannot select or change the route.
- Agentic AI is not allowed to autonomously route, diagnose, prescribe, or contact services. Future agent assistance may be used only for governed coordinator workflows such as service-data update validation and draft content preparation.

## 1.6 Product Strategy, Delivery and Technology Approach

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Delivery Model: Internal Ownership and Selective Outsourcing NetraJyoti will retain product strategy, Bengali conversation design, safety rules, referral logic, privacy, quality assurance, and core application development internally. These are the elements that determine user trust and safety. We will selectively outsource commodity or specialist support such as cloud penetration testing, speech infrastructure, translation validation, and pilot recruitment. External vendors must not control clinical guidance or routing decisions.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <p>Organisational Capabilities and Technology Choices NetraJyoti has full-stack web, Bengali-first UX/content, API and data-modelling capability, and Java unit-test capability. The MVP uses a React + TypeScript PWA, Java 21 + Spring Boot REST APIs, local PostgreSQL, Spring Security configuration, and local Ollama for an optional guarded explanation. Frontend and end-to-end test tooling is available but test evidence is not yet implemented in the repository.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <p>MVP Feature Scope The MVP includes Bengali consent, fixed symptom choices, optional browser speech input with text fallback, urgent warning-sign check, deterministic URGENT/ROUTINE/HUMAN SUPPORT routing, fixed Bengali guidance, district/town or block demo service direction, PHC/ASHA support, caregiver share/copy, and demo-labelled directory content. Optional local-Ollama route explanation is off by default. Feedback, analytics, live directory management, and pilot operations are future scope.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <p>Feature Prioritisation Used MoSCoW. Must: consent, fixed safety routing, urgent/routine/human guidance, Bengali fallback, human support, demo service direction, and safe family sharing. Should: browser speech input and guarded local-Ollama route explanation. Could: live partner integrations, offline cache, reminders, appointment support, feedback analytics, and AI caregiver summary. Will not have in MVP: diagnosis, prescriptions, payment, medical records, autonomous booking, or result-page audio playback.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <p>Pre-Pilot Testing and Release Readiness Required before pilot: API integration, frontend journey, permission-denial, poor-connectivity, browser speech, share/copy, data-flow boundary, and AI failure-mode tests; qualified clinical/content review; Bengali usability and teach-back sessions; accessibility, service-data, privacy, and release-gate evidence. Current repository evidence is limited to Java routing and output-policy unit tests.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <p>AI, GenAI, and Agentic AI Integration</p> <p>The MVP uses browser speech input where supported; it does not offer result-page audio playback. Java safety routing is fully deterministic: it uses matching PostgreSQL criteria and a fixed priority order to choose URGENT, HUMAN SUPPORT, or ROUTINE before any model call.</p> <p>PostgreSQL retrieves route-specific Bengali guidance for the fixed route. Optional local Ollama may create a route explanation from demo-simulated PostgreSQL context. Java policy validation rejects unsafe output and returns fixed Bengali fallback copy. Current seeded content is DEMO_SIMULATED_NOT_FOR_CLINICAL_USE and requires qualified approval before pilot or production use. No patient-identifiable information is sent to Ollama.</p> <p>Agentic AI, authenticated directory management, live service updates, feedback analytics, and AI caregiver summarisation are future scope only.</p> <p><b>Safety routing is fixed before any AI capability is used.</b></p> <p><b>Optional GenAI may explain the fixed route only; it cannot diagnose, prescribe, change urgency, or select a service. Failures show approved Bengali fallback content.</b></p> <p><b>Agentic AI and live directory management remain future scope only.</b></p> |

## 2. Experimental task A: five JTBD interviews

### Required evidence structure

Conduct five real interviews based on the main problem statement. Use a JTBD-style script that explores a real recent eye-care decision, current workaround, anxiety/friction, desired outcome, trusted information source, and reasons for delay. Record the interviews only with participant consent.

| Participant | Exact quote                            | Insight                                   | Feature mapping                                                         | Recording link                                                                                                                                                                            |
|-------------|----------------------------------------|-------------------------------------------|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P1          | "I don't know if this is serious"      | Users need safe urgency clarity           | Deterministic urgent/routine/human-support routing                      | <a href="https://drive.google.com/file/d/1b7s9hYg9FeA5tmZaSeYt5VlzAJmELrFk/view?usp=drive_link">https://drive.google.com/file/d/1b7s9hYg9FeA5tmZaSeYt5VlzAJmELrFk/view?usp=drive_link</a> |
| P2          | "I can't type/read English"            | Language and literacy block access        | Bengali voice input, recognised-text review, text/fixed-choice fallback | <a href="https://drive.google.com/file/d/1Xcj8n3xReNfcPQ4wZW-uPEcSlmUlfUUQ/view?usp=drive_link">https://drive.google.com/file/d/1Xcj8n3xReNfcPQ4wZW-uPEcSlmUlfUUQ/view?usp=drive_link</a> |
| P3          | "I don't know where to go"             | Guidance must become practical navigation | District/block service direction, PHC/ASHA options                      | <a href="https://drive.google.com/file/d/1pg8SybVlhFRRh3HVhUqw6e96TDsaFGxd/view?usp=drive_link">https://drive.google.com/file/d/1pg8SybVlhFRRh3HVhUqw6e96TDsaFGxd/view?usp=drive_link</a> |
| P4          | "I need to ask other family members"   | Care decisions are shared                 | Caregiver message, browser share/copy fallback                          | <a href="https://drive.google.com/file/d/12f8oSr1wpeM1OH-5R4F1o05B9FfuP7t0/view?usp=drive_link">https://drive.google.com/file/d/12f8oSr1wpeM1OH-5R4F1o05B9FfuP7t0/view?usp=drive_link</a> |
| P5          | "I'm not comfortable using this alone" | Human reassurance is essential            | ASHA/PHC/eye-clinic/trusted-family human-support path                   | <a href="https://drive.google.com/file/d/1_IKpLxz2mwhgLP_D_2WxUBTs-51eJTdU/view?usp=drive_link">https://drive.google.com/file/d/1_IKpLxz2mwhgLP_D_2WxUBTs-51eJTdU/view?usp=drive_link</a> |

## 3. Delivery, governance, Jira, and Lean Canvas

### 3.1 Methodology and project plan

NetraJyoti uses Lean UX plus Agile Kanban. Lean UX reduces product uncertainty through real user evidence and rapid Bengali prototype tests. The Kanban board visualises the flow of reviewed requirements into testable increments, while safety and content review act as defined release gates.

| Phase                        | Weeks | Deliverable / gate                                   |
|------------------------------|-------|------------------------------------------------------|
| Discovery and evidence       | 1-2   | JTBD interviews, personas, Lean Canvas, hypotheses.  |
| Journey and prototype        | 3-4   | User flow, bilingual wireframes, usability findings. |
| Technical design and backlog | 5     | PRD, HLD, Jira stories, estimates, safety rules.     |

| Phase                     | Weeks | Deliverable / gate                                                                             |
|---------------------------|-------|------------------------------------------------------------------------------------------------|
| MVP build                 | 6-9   | Core routing; urgent/routine/human routes; service direction; sharing; controlled AI/fallback. |
| Pilot readiness           | 10    | Privacy, content, service-data, and release checks.                                            |
| Validation and final pack | 11-12 | Usability/pilot insight, refinements, final artefacts and demo.                                |

3.2 Jira delivery evidence

Epic: [NET-1 - NetraJyoti Epic MVP](#)

Spaces / NetraJyoti / NET-1

**Epic: NetraJyoti AI — Bengali Voice-First Eye-Health Guidance MVP**

Add a child work item

Link work item

Create

...

Key details

Rahul Kar raised this request via Jira

Description

Problem

Rural Bengali-speaking users may delay eye-care decisions because they are unsure how serious a concern is, where to go, and which advice to trust. Text-heavy or English-first services create additional barriers.

Feature Goal

Provide a safe Bengali voice-first experience that helps users understand the appropriate next step and receive reliable service direction. NetraJyoti is guidance and referral support—not a diagnostic tool—and users must verify service availability locally before travel.

Target Users

- Rural Bengali-speaking users
- Caregivers
- ASHA/community health workers
- Eye-care outreach coordinators

Kanban board: [Board 68](#)

NetraJyoti AI MVP Delivery Board

Kanban board

Reports

+

Search board

Version

Epic

Label

Quick filters

Group: Query

TO DO

IN PROGRESS

TESTING / CONTENT REVIEW

DONE 11

Story: AI-assisted caregiver summary from approved result content

Epic: NetraJyoti AI — Bengali Voice-First Eye-Health Guidance ...

☒ NET-12

5

Story: AI-assisted plain-Bengali explanation of approved guidance

Epic: NetraJyoti AI — Bengali Voice-First Eye-Health Guidance ...

☒ NET-11

5

Story: Feedback and privacy-safe pilot analytics

Epic: NetraJyoti AI — Bengali Voice-First Eye-Health Guidance ...

☒ NET-10

10

Story: Safety, accessibility and fallback quality assurance

Epic: NetraJyoti AI — Bengali Voice-First Eye-Health Guidance ...

☒ NET-9

10

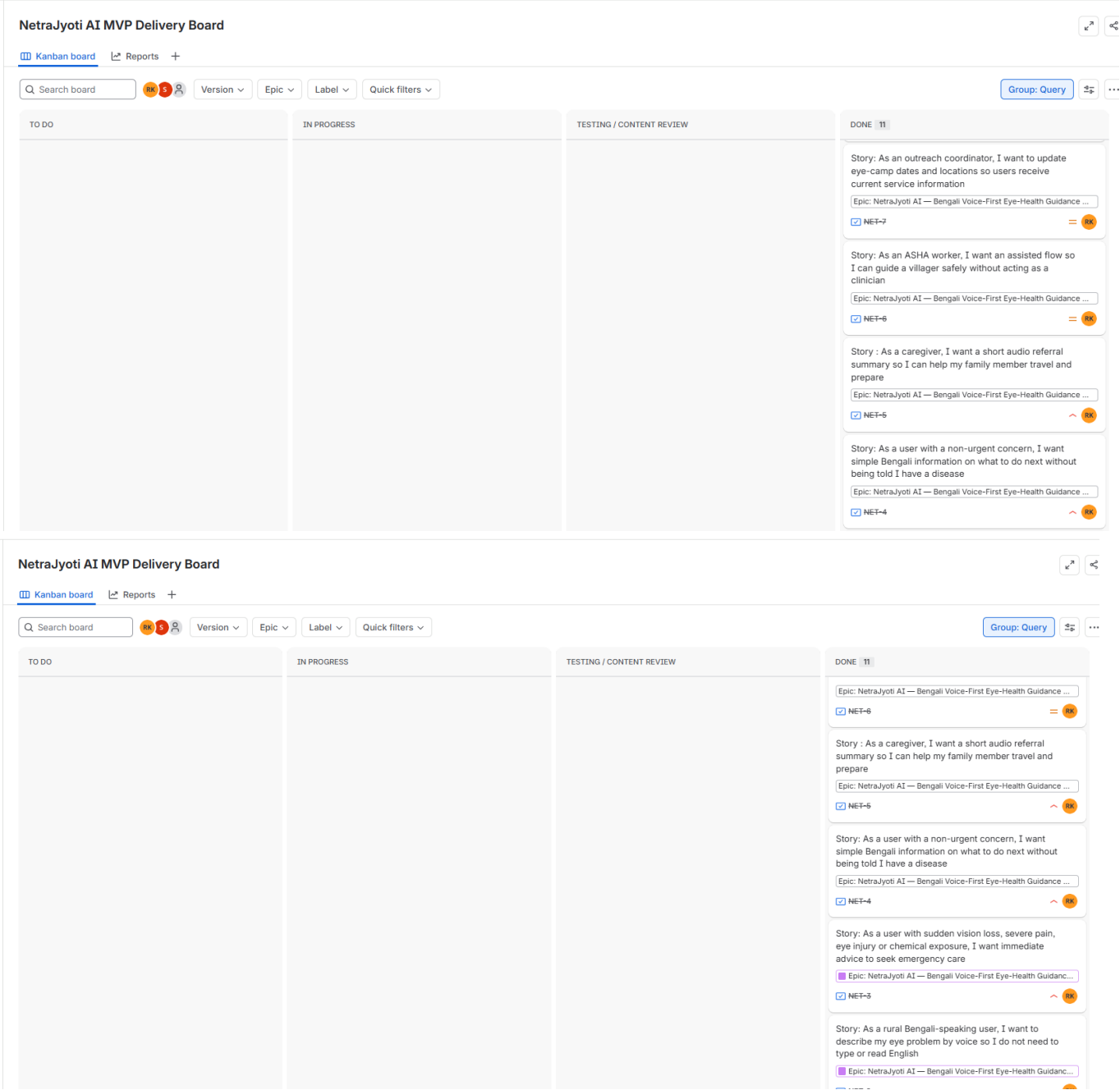
Story: Consent, privacy and minimum-data handling

Epic: NetraJyoti AI — Bengali Voice-First Eye-Health Guidance ...

☒ NET-8

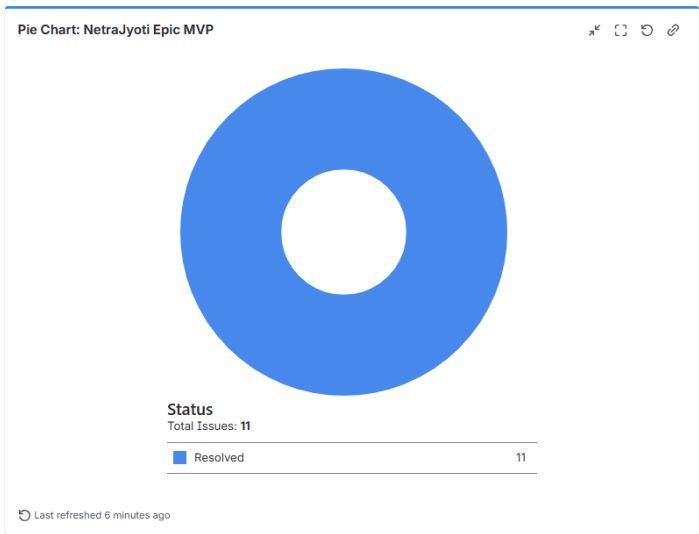
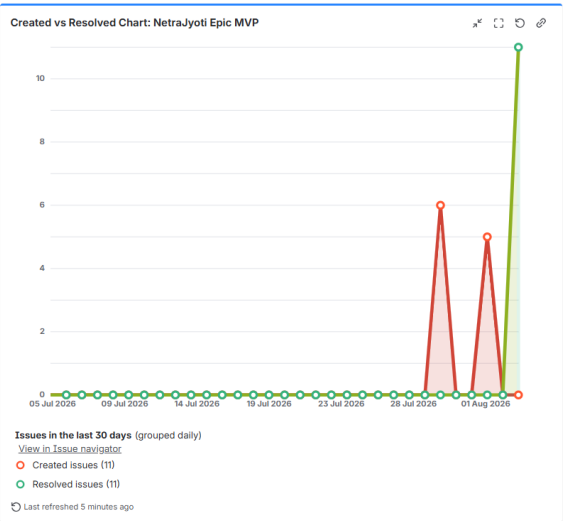
10

Story: As an outreach coordinator, I want to update



Historical dashboard snapshot: Dashboard 10035 (superseded). Current dashboard: [NetraJyoti AI MVP Dashboard \(ID 10001\)](#)

NetraJyoti AI MVP Dashboard V1



Filter Results: NetraJyoti Epic MVP

| T                                   | Key    | Summary                                                                                                                                | P |
|-------------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------|---|
| <input checked="" type="checkbox"/> | NET-12 | Story: AI-assisted caregiver summary from approved result content                                                                      | = |
| <input checked="" type="checkbox"/> | NET-11 | Story: AI-assisted plain-Bengali explanation of approved guidance                                                                      | = |
| <input checked="" type="checkbox"/> | NET-10 | Story: Feedback and privacy-safe pilot analytics                                                                                       | = |
| <input checked="" type="checkbox"/> | NET-9  | Story: Safety, accessibility and fallback quality assurance                                                                            | = |
| <input checked="" type="checkbox"/> | NET-8  | Story: Consent, privacy and minimum-data handling                                                                                      | = |
| <input checked="" type="checkbox"/> | NET-7  | Story: As an outreach coordinator, I want to update eye-camp dates and locations so users receive current service information          | = |
| <input checked="" type="checkbox"/> | NET-6  | Story: As an ASHA worker, I want an assisted flow so I can guide a villager safely without acting as a clinician                       | = |
| <input checked="" type="checkbox"/> | NET-5  | Story : As a caregiver, I want a short audio referral summary so I can help my family member travel and prepare                        | ^ |
| <input checked="" type="checkbox"/> | NET-4  | Story: As a user with a non-urgent concern, I want simple Bengali information on what to do next without being told I have a disease   | ^ |
| <input checked="" type="checkbox"/> | NET-3  | Story: As a user with sudden vision loss, severe pain, eye injury or chemical exposure, I want immediate advice to seek emergency care | ^ |

1-10 of 11

Last refreshed 7 minutes ago

3.3 Ethical considerations

| Area          | Risk                                                                    | Safeguard                                                                                                                                                                                                                                   |
|---------------|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Non-diagnosis | False clinical confidence.                                              | Clear purpose; no disease label, prescription, or clinician-replacement claim.                                                                                                                                                              |
| Urgency       | Delay or inappropriate escalation.                                      | Java uses PostgreSQL criteria and deterministic priority to fix every route before optional local Ollama explanation. Java validates output and falls back to fixed Bengali copy. Clinical approval is required before pilot or production. |
| AI            | Invented advice or changed urgency.                                     | Backend-only feature flag; post-route only; schema/policy validation; fixed Bengali fallback.                                                                                                                                               |
| Privacy       | Exposure of name, phone, address, voice, or location.                   | Data minimisation, explicit consent, client-side temporary details, no PII logs by default.                                                                                                                                                 |
| Inclusion     | Language, literacy, device, age, disability, or connectivity exclusion. | Static demo facility cards are visibly labelled and users are told to confirm availability. Authenticated partner updates, live verification, and freshness operations are future pilot capabilities.                                       |
| Service data  | Incorrect camp/clinic information.                                      | Static demo labels; confirm before travel; no unsupported claims. Verification, update, and freshness controls are future pilot scope.                                                                                                      |

### 3.4 Lean Canvas

| Block                    | NetraJyoti response                                                                                                         |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Problem                  | Confusion, delay, mistrust, and service uncertainty before eye care.                                                        |
| Customer segments        | Rural Bengali-speaking users; caregivers; ASHA/community workers; outreach coordinators.                                    |
| Unique value proposition | Bengali-first safe next-step guidance for eye problems.                                                                     |
| Solution                 | Reviewed safety routing, service direction, human support, and family sharing.                                              |
| Channels                 | Eye-care partners, eye camps, ASHA workers, QR/posters, caregiver sharing, and local outreach.                              |
| Costs                    | Product build, cloud/data, clinical/content review, service verification, usability research, and optional AI.              |
| Key metrics              | Journey completion, route distribution, service-direction use, comprehension, share rate, data freshness, AI fallback rate. |
| Unfair advantage         | Local partner trust, reviewed Bengali content, safety workflow, and curated service information.                            |

## 4. STP, brand, 7Ms, and launch plan

### 4.1 STP and brand

| Element       | NetraJyoti plan                                                                                                                                  |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Segmentation  | Rural/near-rural Bengali speakers; eye-care uncertainty level; caregiver role; digital confidence; proximity to vision/eye-care services.        |
| Targeting     | Primary: Bengali-speaking adults and caregivers in rural West Bengal with a current eye concern and uncertainty about the appropriate next step. |
| Positioning   | NetraJyoti helps people understand a safe next step and the suitable path to eye care in Bengali. It does not diagnose disease.                  |
| Brand         | Name: NetraJyoti AI. Meaning: light/clarity for the eyes. Personality: calm, respectful, practical, trustworthy, Bengali-first.                  |
| Brand promise | Clear, safe next steps for eye-care decisions, in the user's language.                                                                           |

## 4.2 7Ms for the new product

| M           | Decision                                                                                                                       |
|-------------|--------------------------------------------------------------------------------------------------------------------------------|
| Mission     | Reduce delay, fear, and confusion before appropriate eye care.                                                                 |
| Market      | Rural West Bengal communities reached through care partners and community-health networks.                                     |
| Message     | Bengali-first safe next-step guidance; not a diagnosis.                                                                        |
| Media       | ASHA/partner referrals, eye-camp posters, QR cards, WhatsApp family sharing, local outreach, and partner channels.             |
| Money       | Pilot funded through CSR, NGO, eye-care partners, public-health grants, or implementation agreements rather than user payment. |
| Manpower    | Product, engineering, QA, clinical/content reviewer, outreach coordinator, partner health workers.                             |
| Measurement | Completion, route/service use, comprehension, verified data freshness, pilot feedback, and safety/fallback signals.            |

## 4.3 End-to-end launch plan

| Launch phase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Actions                                                                                                                          | Exit gate                                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| Prepare                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Approve safety/content, verify service data, train partner/ASHA referrals, create QR/outreach materials, test live MVP.          | Clinical, privacy, usability, and operational approval. |
| Pilot launch                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Launch with one or two partner areas; use assisted onboarding at camps/vision centres; capture non-identifying journey evidence. | No critical safety or comprehension failure.            |
| Learn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Review friction, route distributions, data freshness, share use, and interviews weekly; prioritise fixes.                        | Evidence supports a wider pilot.                        |
| Scale carefully                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Expand partner locations after service-data ownership and support model are proven.                                              | Operational readiness and documented impact hypothesis. |
| <p>60-second launch pitch. NetraJyoti AI is a Bengali-first eye-care guidance service for people who are unsure what to do when an eye problem occurs. It does not diagnose disease. Instead, it helps users identify warning signs, understand a safe next step, and find the right eye-care route such as a vision centre, clinic, hospital, PHC, or trusted health worker. We will launch through eye-care partners, ASHA/community workers, eye camps, and simple mobile access so guidance reaches people before uncertainty becomes delay.</p> |                                                                                                                                  |                                                         |

## 5. Channel strategy, user flow, and channel experiment

### 5.1 Omnichannel strategy

NetraJyoti should be omnichannel, not purely online. A mobile web/PWA experience provides scalable, low-friction access, while eye camps, vision centres, ASHA workers, PHCs, and trusted caregivers provide human reassurance and support for people with low digital confidence or limited connectivity.

| Channel                       | Role                                                                   | Owned / partner                         | Parity control                                                                                      |
|-------------------------------|------------------------------------------------------------------------|-----------------------------------------|-----------------------------------------------------------------------------------------------------|
| Mobile PWA                    | Self-service symptom, route, service direction, family message.        | Owned digital experience.               | Same demo safety content and fixed route logic. Live service-directory data parity is future scope. |
| ASHA / PHC / eye-care partner | Assisted onboarding, human escalation, referral, service confirmation. | Partner/franchised-like local delivery. | Training, approved scripts, referral standards, data-update process.                                |
| Eye camps / vision centres    | Physical awareness, QR entry, service follow-through, feedback.        | Partner channel.                        | Consistent brand, printed scripts, verified service details.                                        |
| Family sharing / WhatsApp     | Caregiver involvement and urgency communication.                       | User-owned social channel.              | Approved share template retains safe next step and selected symptoms.                               |

### NetraJyoti AI - Two-Channel Experiment

#### 1. Objective

Compare the same non-diagnostic eye-care enquiry in two channels to identify completion time, friction, drop-offs, trust leakage, and scaling constraints.

#### 2. Test scenario

A Bengali-speaking caregiver wants guidance for an older family member with red, itchy, watering eyes and is unsure whether to go to a vision centre, PHC, or eye clinic.

**Success criterion: the caregiver reaches a safe routine-care recommendation and receives a shareable care summary or assisted referral direction without being shown a disease diagnosis.**

#### 3. Channels tested

| Channel                  | Real action attempted                                                                                                                                         | Test setup                                                   |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| PWA self-service         | Open NetraJyoti, give consent, select symptoms, complete the warning-sign check, select district/town, view routine guidance, and create a caregiver summary. | Android phone, mobile data, Bengali-first UI.                |
| Assisted partner enquiry | Ask an ASHA worker or eye-camp outreach coordinator for the same next-step guidance and nearby service direction.                                             | In-person or phone-assisted enquiry using the same scenario. |

#### 4. Observation log

| Measure           | PWA self-service       | Assisted partner enquiry    |
|-------------------|------------------------|-----------------------------|
| Participant label | PWA-01 (caregiver, 38) | Assisted-01 (caregiver, 41) |

| Measure                           | PWA self-service                                                | Assisted partner enquiry                                                        |
|-----------------------------------|-----------------------------------------------------------------|---------------------------------------------------------------------------------|
| Time to begin                     | 25 seconds                                                      | 2 minutes 10 seconds                                                            |
| Time to safe next step            | 3 minutes 42 seconds                                            | 8 minutes 35 seconds                                                            |
| Time to share / service direction | 4 minutes 15 seconds                                            | 10 minutes 05 seconds                                                           |
| Outcome                           | Routine eye-check recommendation and shareable summary created. | PHC / vision-centre direction explained verbally; no written summary initially. |
| Completion                        | Completed independently after one prompt.                       | Completed with staff availability.                                              |

## 5. Friction points and drop-off moments

| Channel  | Observed friction                                                         | Potential drop-off                                      | Recommended improvement                                                      |
|----------|---------------------------------------------------------------------------|---------------------------------------------------------|------------------------------------------------------------------------------|
| PWA      | Consent instruction was read but the checkbox was initially missed.       | User may leave before starting.                         | Place a short Bengali instruction directly above a large consent control.    |
| PWA      | District/town selection depends on familiarity with administrative names. | User may exit when locality is not easily recognised.   | Add searchable Bengali locality names, recent choices, and support fallback. |
| PWA      | User hesitated before allowing location access.                           | User may reject permission and abandon service search.  | Explain location is optional; offer district/town selection first.           |
| Assisted | Worker was not immediately available.                                     | User may delay until next visit or give up.             | Provide QR/PWA self-service and escalation contact path.                     |
| Assisted | Guidance was verbal and details were not retained.                        | Family may forget suggested service or urgency message. | Generate a Bengali shareable summary the worker can send or read aloud.      |

## 6. Trust leakage and scaling constraints

| Area            | Finding                                                                   | Why it matters                                                     | Mitigation                                                                                                     |
|-----------------|---------------------------------------------------------------------------|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| PWA trust       | User questioned whether an app can understand an eye problem.             | Fear of an incorrect diagnosis can prevent use.                    | Explain value positively: safe next steps and a suitable eye-care route; keep non-diagnostic boundary concise. |
| Directory trust | Service listings may look unverified or outdated.                         | A wrong location damages trust quickly.                            | Show verification date and service type; label demo data; use coordinator update workflow.                     |
| Assisted trust  | User depends on a worker's availability or memory.                        | Experience varies by person and day.                               | Use standard guidance cards and shareable summaries linked to reviewed rules.                                  |
| PWA scale       | Access, connectivity, voice/browser support, and digital confidence vary. | Low marginal cost but needs assisted onboarding and simple design. | Use an offline-friendly PWA and local-language support.                                                        |
| Assisted scale  | ASHA/outreach availability and training are limited.                      | High trust but capacity grows slowly.                              | Use scripts, referral workflow, and service-directory governance.                                              |

## 7. Conclusion

The PWA route is faster when the user can operate a phone comfortably and identify their area. The assisted route creates more human trust and supports low-confidence users, but it is slower and depends on worker availability.

## 5.2 User flow and journey

### NetraJyoti AI — simple user journey flow

Reviewed safety rules determine the route. Optional AI never determines urgency.

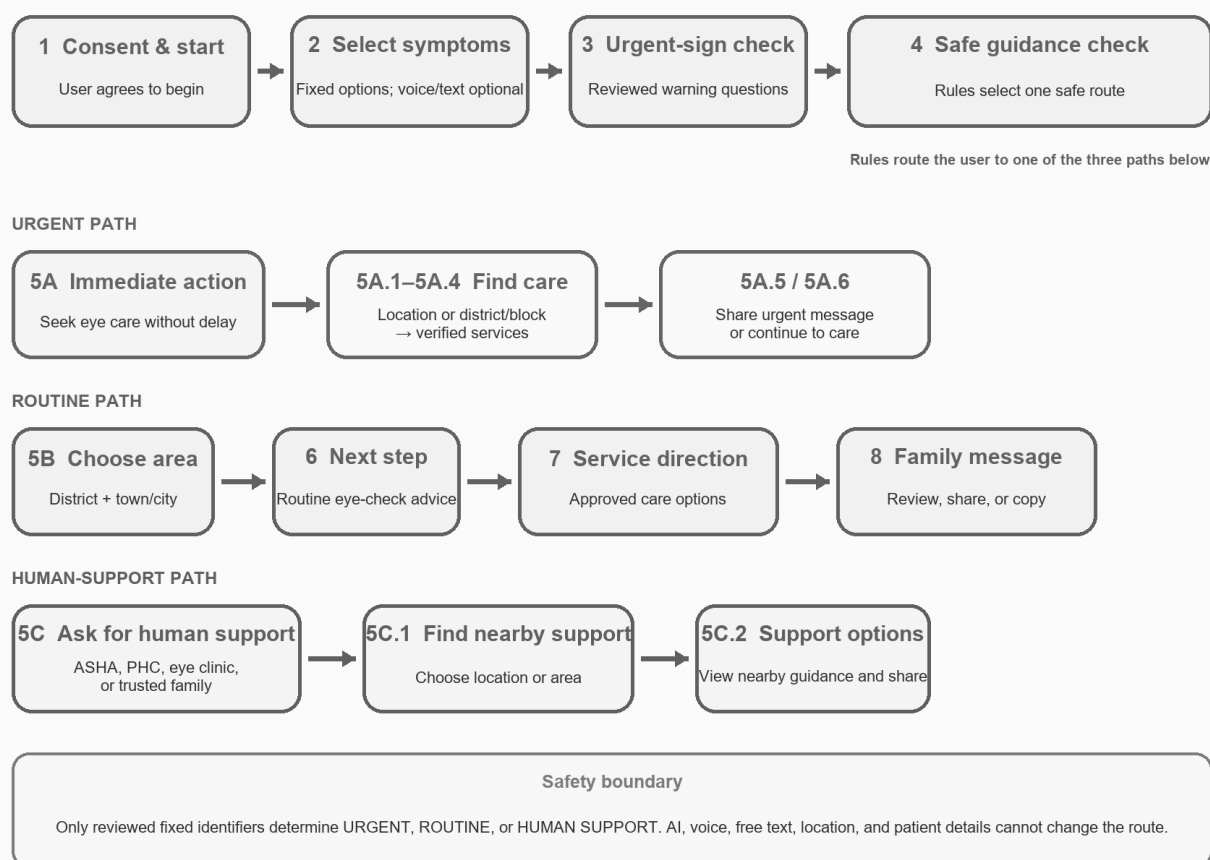


Figure 1. NetraJyoti AI end-to-end journey: core intake followed by urgent, routine, or human-support routes.

## 6. Retention, CLV, customer service, and growth

### 6.1 Retention and renewal research

| Participant | Verbatim renewal objection                                                                               | Emotional driver        | Retention lever                                                                | Video link                                                                                                                                                                                  |
|-------------|----------------------------------------------------------------------------------------------------------|-------------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R1          | Concern that the product may not be trusted for an eye-care decision                                     | Trust                   | Show clinician-reviewed safety boundary; add visible human-support option      | <a href="https://drive.google.com/file/d/1iLTcu5Fd_LUWqA2mkzf6HhJC7-GhIAcF/view?usp=drive_link">https://drive.google.com/file/d/1iLTcu5Fd_LUWqA2mkzf6HhJC7-GhIAcF/view?usp=drive_link</a>   |
| R2          | Concern that the process may be difficult for users with low digital confidence                          | Complexity              | Bengali voice/text fallback, simple steps, and ASHA/PHC-assisted support       | <a href="https://drive.google.com/file/d/1TuiJbZ_ZcpFi6-1wh647SC5LyVa30hs93/view?usp=drive_link">https://drive.google.com/file/d/1TuiJbZ_ZcpFi6-1wh647SC5LyVa30hs93/view?usp=drive_link</a> |
| R3          | Concern that the product may not be useful after the first guidance session                              | Low perceived value     | Onboarding explaining repeat-use situations; short follow-up education/support | <a href="https://drive.google.com/file/d/1tJGtZTGYdTdY1-jYok5-AUs8xDKAHOWc/view?usp=drive_link">https://drive.google.com/file/d/1tJGtZTGYdTdY1-jYok5-AUs8xDKAHOWc/view?usp=drive_link</a>   |
| R4          | Concern about accessibility of the app                                                                   | Complexity, convenience | Simple steps, ASHA/PHC-assisted support                                        | <a href="https://drive.google.com/file/d/1nqYhq4s_fz9oW9inAqAzLOEcVWDJLEVG/view?usp=drive_link">https://drive.google.com/file/d/1nqYhq4s_fz9oW9inAqAzLOEcVWDJLEVG/view?usp=drive_link</a>   |
| R5          | Concern that the product might not be trusted and should incorporate notification and follow up messages | Trust                   | More information, human support option                                         | <a href="https://drive.google.com/file/d/1IGCzljpkql7rkOwn8MGc25Drry_yAh1L/view?usp=drive_link">https://drive.google.com/file/d/1IGCzljpkql7rkOwn8MGc25Drry_yAh1L/view?usp=drive_link</a>   |

## 6.2 Customer-service and CLV strategy

| Area                          | Approach                                                                                                                                                                                     |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Customer support              | In-app guidance, family-share guidance, ASHA/PHC/clinic escalation, and a service-data disclaimer. A feedback route is future scope. No automated clinical diagnosis.                        |
| Increase trust and repeat use | Keep content short and understandable; show safe next step; label service direction as demo; make caregiver support simple; offer human fallback.                                            |
| CLV / renewal                 | For partner-funded model: demonstrate value through completed care-navigation journeys, service-direction use, community reach, and partner trust rather than consumer subscription revenue. |
| Growth loops                  | Caregiver share loop and human-support referral guidance are current. Partner update, analytics, and feedback learning loops are future pilot scope.                                         |
| Habit building                | Use only ethically appropriate reminders/support after explicit consent. Avoid dark patterns, fear-based nudges, and health claims.                                                          |

## 7. Experimental task C: business-owner support research

| Business-owner support research: retain evaluator-accessible recordings and document five customer-support queries, best/worst resolution times, follow-ups, escalations, drop-offs, and specific frustration moments before final submission. |                                                                                                      |                                                                                             |                                                                                                                                                            |                                                                                                                                                                                           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Owner / industry                                                                                                                                                                                                                               | Frequent issue / friction                                                                            | Current resolution                                                                          | Improvement suggestion                                                                                                                                     | Recording link                                                                                                                                                                            |
| B1(Cloud Kitchen Owner)                                                                                                                                                                                                                        | Order delays during peak hours and incorrect orders affecting customer ratings.                      | Kitchen staff manually prioritize orders and resolve complaints through delivery platforms. | Implement order management software with real-time kitchen queue tracking, standardized preparation workflows, and packaging verification before dispatch. | <a href="https://drive.google.com/file/d/1HDBlyE2KTctfrqzGaYwRXxFR5i88DTJ6/view?usp=drive_link">https://drive.google.com/file/d/1HDBlyE2KTctfrqzGaYwRXxFR5i88DTJ6/view?usp=drive_link</a> |
| B2 (Caregiver Center owner)                                                                                                                                                                                                                    | Difficulty managing schedules, medication reminders, and communication with multiple family members. | Uses paper notes, phone calls, and messaging apps to coordinate care.                       | Use a shared caregiving app for schedules, medication tracking, task updates, and family notifications to reduce missed tasks and improve transparency.    | <a href="https://drive.google.com/file/d/1W5Of5-O0O5hWXY37TjplUDdTIAVvslqg/view?usp=drive_link">https://drive.google.com/file/d/1W5Of5-O0O5hWXY37TjplUDdTIAVvslqg/view?usp=drive_link</a> |
| B3(Fruit seller)                                                                                                                                                                                                                               | Unsold fruits spoil quickly, leading to daily inventory loss and inconsistent income.                | Seller reduces prices at the end of the day or absorbs the loss personally.                 | Introduce simple demand forecasting, digital inventory tracking, and partnerships with nearby shops or food vendors to sell excess stock before spoilage.  | <a href="https://drive.google.com/file/d/144bs6sOx2cgXxvU81dLSTPn2AU1iWqH/view?usp=drive_link">https://drive.google.com/file/d/144bs6sOx2cgXxvU81dLSTPn2AU1iWqH/view?usp=drive_link</a>   |

### Application to Netrajyoti: expected support design

- Provide clear route explanations and visible next-step actions instead of generic advice.
- Offer human support and caregiver/family sharing alongside self-service guidance.
- Clearly state that service directions are guidance only; users should confirm availability before travelling.
- Track non-identifying friction signals and verify service information regularly.
- Use simple, respectful Bengali and provide a clear restart or return option at every major point in the journey.

## 8. Jira Epic backlog traceability

| Current backlog status. NET-1 is In Progress. The Epic has 11 linked delivery stories (NET-2 to NET-12). All 11 stories are Resolved in the current Jira export, with 51 total story points. This is the source of truth for the delivery appendix. |                                                                                                 |                            |                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|----------------------------|-------------------------------------------------------------------|
| Jira story                                                                                                                                                                                                                                          | Backlog scope                                                                                   | Priority / status / points | PRD and journey traceability                                      |
| NET-2                                                                                                                                                                                                                                               | Bengali voice capture with recognised-text review and text fallback.                            | High / Resolved / 5        | Voice-first access; Step 2 symptom input; accessibility fallback. |
| NET-3                                                                                                                                                                                                                                               | Immediate urgent-care advice for sudden vision loss, severe pain, injury, or chemical exposure. | High / Resolved / 5        | Step 3 red-flag check; Step 4 urgent route; Step 5A branch.       |
| NET-4                                                                                                                                                                                                                                               | Plain Bengali, non-diagnostic next-step guidance for non-urgent concerns.                       | High / Resolved / 5        | Routine outcome; Steps 5B, 6, 7, and 8.                           |

|        |                                                                                                                                         |                       |                                                                                                                  |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------------|------------------------------------------------------------------------------------------------------------------|
| NET-5  | Caregiver prepared message with browser share/copy. Result-page audio/referral playback is not in the current MVP.                      | High / Resolved / 5   | Family-sharing flow; urgent 5A.5 and routine Step 8, with no result-page audio playback.                         |
| NET-6  | ASHA-assisted journey for safely supporting a villager.                                                                                 | Medium / Resolved / 3 | Human-support path; Steps 5C, 5C.1, and 5C.2.                                                                    |
| NET-7  | Future pilot: authenticated outreach-coordinator management of eye-camp dates and location information. The MVP uses static demo cards. | Medium / Resolved / 5 | Future service-directory governance and data freshness operations; current MVP shows demo service direction.     |
| NET-8  | Consent, privacy, and minimum-data handling.                                                                                            | Medium / Resolved / 5 | Step 1 consent; optional details/share consent; data boundaries.                                                 |
| NET-9  | Safety, accessibility, and fallback quality assurance.                                                                                  | Medium / Resolved / 5 | Deterministic rules, tested fallbacks, usability/release gates.                                                  |
| NET-10 | Future pilot: privacy-safe feedback and non-content analytics. These are not implemented in the current MVP.                            | Medium / Resolved / 5 | Future pilot metrics and operational learning; not a current MVP capability.                                     |
| NET-11 | Controlled AI plain-Bengali explanation of approved guidance.                                                                           | Medium / Resolved / 5 | Server feature flag; rules-first route; validation and fixed-copy fallback.                                      |
| NET-12 | Future capability: AI caregiver summary from approved result content. Current Step 8 uses a fixed prepared share message.               | Medium / Resolved / 3 | Future AI caregiver-summary scope; current Step 8 shares fixed content and sends no user-entered data to Ollama. |

## Evidence and Supporting Links

- High\_Level\_Design\_Document: [docs/project-documents/NetraJyoti\\_High\\_Level\\_Design.pdf](#)
- Technical\_PRD.Document: [docs/project-documents/NetraJyoti\\_Technical\\_PRD.pdf](#)
- PRD: [docs/project-documents/NetraJyoti\\_MVP\\_PRD.pdf](#)
- Project Plan: [docs/project-documents/NetraJyoti\\_Project\\_Plan.pdf](#)
- NetraJyoti wireframe: <https://rahulkarcj.github.io/netrajyoti-ai-mvp/>
- GitHub repository: <https://github.com/rahulkarcj/netrajyoti-ai-mvp>
- Jira Epic: <https://rkarpmdemo.atlassian.net/browse/NET-1>
- Jira dashboard: <https://rkarpmdemo.atlassian.net/jira/dashboards/10001>
- Jira Kanban board: <https://rkarpmdemo.atlassian.net/jira/people/557058:3da47173-80fe-4c5f-9db8-804c2541a6c8/boards/68>
- MVPDemoRecording: [https://drive.google.com/drive/folders/1nOJ4W02Vaz4ZS5RFRv\\_ljrkiAqF0jnJA?usp=drive\\_link](https://drive.google.com/drive/folders/1nOJ4W02Vaz4ZS5RFRv_ljrkiAqF0jnJA?usp=drive_link)